Unfolded Hyperspectral and Multispectral Image Fusion with Exact Observation Operators

Supplementary Material

Anonymous CVPR submission

Equations, theorems, sections and tables of the main paper are referred to by their numbers there (*e.g.* Theorem 1, Tab. 1); items of this supplement carry the prefix S.

S1. The exact adjoint of reflect padding

Every gradient step on $\|\mathcal{D}\mathbf{X} - \mathbf{Y}_\mathrm{H}\|^2$ needs $\mathcal{D}^\top = \mathcal{P}^\top\mathcal{K}^\top\mathcal{S}^\top$, where $\mathcal{P}$ pads by p pixels, $\mathcal{K}$ is a valid correlation with the $(2p+1)\times(2p+1)$ kernel and $\mathcal{S}$ keeps every s-th pixel from index zero. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with output padding $s-1$. The adjoint of reflect padding adds each padded margin back onto the pixels it was copied from. For a padded row index $j \in [0, H+2p)$ the source row is $p-j$ for $j<p$, $j-p$ in the interior and $2H-2+p-j$ in the bottom margin, hence

$$(\mathcal{P}^\top\mathbf{g})_i = \mathbf{g}_{i+p} + [1 \le i \le p]\,\mathbf{g}_{p-i} + [H-1-p \le i \le H-2]\,\mathbf{g}_{2H-2+p-i}, \tag{S1}$$

applied to rows and then to columns. With (S1), the operator in the network is exactly the simulation operator, and $\langle\mathcal{D}\mathbf{X}, \mathbf{Y}\rangle = \langle\mathbf{X}, \mathcal{D}^\top\mathbf{Y}\rangle$ holds to a relative error below 10^{-13} in double precision at every scale factor. $\mathcal{R}^\top$ is the transpose of the response matrix.

S2. Proofs

Theorem 1. (a) By (4), $\mathcal{D}_\theta^\top\mathcal{D}_\theta$ applies $\mathbf{D}_\mathrm{r}^\top\mathbf{D}_\mathrm{r}$ and $\mathbf{D}_\mathrm{c}^\top\mathbf{D}_\mathrm{c}$ to the two spatial modes and the identity to the spectral mode, and $\mathcal{R}_\phi^\top\mathcal{R}_\phi$ applies $\mathbf{r}_\phi^\top\mathbf{r}_\phi$ to the spectral mode alone. Each factor is diagonalized by the corresponding factor of $\mathcal{Q}$, and an identity factor by any orthonormal basis. (b) In this basis the matrix of (2) is diagonal with entries $1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\beta_k\nu_b > 0$. (c) $\mathcal{N}(\mathcal{D}_\theta)\cap\mathcal{N}(\mathcal{R}_\phi) = \mathcal{N}(\mathcal{D}_\theta^\top\mathcal{D}_\theta + \mathcal{R}_\phi^\top\mathcal{R}_\phi)$, whose eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\nu_b$ are sums of nonnegative terms and vanish only when both terms do. The basis tensors with $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} > 0$ span the row space of $\mathbf{D}_\mathrm{r}\otimes\mathbf{D}_\mathrm{c}$, of dimension hw, and those with $\nu_b > 0$ the row space of $\mathbf{r}_\phi$, of dimension M. The orthogonal projector onto the null space is therefore $(\mathbf{I}-\mathbf{P}_\mathrm{s})\otimes(\mathbf{I}-\mathbf{P}_\mathrm{r}\otimes\mathbf{P}_\mathrm{c})$ in mode order, with $\mathbf{P}_\mathrm{s}, \mathbf{P}_\mathrm{r}, \mathbf{P}_\mathrm{c}$ the projectors onto the row spaces of $\mathbf{r}_\phi, \mathbf{D}_\mathrm{r}, \mathbf{D}_\mathrm{c}$, and its dimension is $(B-M)(HW-hw)$. □

Theorem 2. (a) $\tilde{\mathbf{A}}\mathbf{X}^* - \mathbf{Y} = \mathbf{0}$ makes both updates vanish. (b) With $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$, a direct computation gives $T(\mathbf{X}^*) - \mathbf{X}^* = (\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{e}$ with $\mathbf{e} = \mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$. In the singular value decomposition of $\mathbf{B}$, $(\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ has singular values $b_i/(1+b_i^2) \le \frac{1}{2}$. Similarly $G(\mathbf{X}^*)-\mathbf{X}^* = \mathbf{B}^\top\mathbf{e}$, whose norm is at most $\|\mathbf{B}\|\,\|\mathbf{e}\|$. (c) T_k is nonexpansive, since $\|T_k(\mathbf{V}) - T_k(\mathbf{V}')\| = \|(\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}(\mathbf{V}-\mathbf{V}')\| \le \|\mathbf{V}-\mathbf{V}'\|$. With $e_k = \|\mathbf{X}_k - \mathbf{X}^*\|$ and $(\mathcal{I}+\mathcal{P}_k)(\mathbf{X}^*) = \mathbf{X}^*$,

$$\begin{aligned} e_k &\le L_k\|T_k(\mathbf{X}_{k-1}) - \mathbf{X}^*\| \\ &\le L_k\big(\|T_k(\mathbf{X}_{k-1}) - T_k(\mathbf{X}^*)\| + \|T_k(\mathbf{X}^*) - \mathbf{X}^*\|\big) \le L_k\big(e_{k-1} + \tfrac{1}{2}\|\mathbf{e}_k\| \end{aligned}$$

and unrolling from $e_0 = 0$ gives the bound. The gradient step is not nonexpansive in general, so the bound has no counterpart for it. □

Corollary 1. A low-resolution pixel is a weighted sum of the $(2p+1)\times(2p+1)$ high-resolution pixels around its sampling position. Two border rules give the same value whenever this window lies inside the image, i.e., unless the sampling position is within p pixels of the border; with sampling positions $0, s, 2s, \ldots$ this concerns the first $\lceil p/s\rceil$ low-resolution rows and columns (and, at the far border, those whose window crosses it). □

Theorem 3. (a) On vectorized images, $\mathcal{D}_\theta = \mathbf{I}_B\otimes\bar{\mathcal{D}}_\theta$ and $\mathcal{R}_\phi = \mathbf{r}_\phi\otimes\mathbf{I}_N$, with $\bar{\mathcal{D}}_\theta$ the single-band spatial operator and $N = HW$. By the mixed-product rule, $(\mathbf{I}_M\otimes\bar{\mathcal{D}}_\theta)(\mathbf{r}_\phi\otimes\mathbf{I}_N) = \mathbf{r}_\phi\otimes\bar{\mathcal{D}}_\theta = (\mathbf{r}_\phi\otimes\mathbf{I}_{N/s^2})(\mathbf{I}_B\otimes\bar{\mathcal{D}}_\theta)$. With $\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathbf{X}$ and $\mathbf{Y}_\mathrm{H} = \mathcal{D}_{\theta^*}\mathbf{X}$, $\mathcal{D}_{\theta^*}\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathcal{D}_{\theta^*}\mathbf{X} = \mathcal{R}_{\phi^*}\mathbf{Y}_\mathrm{H}$. (b) By (a), $J(\theta,\phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2$ with $\mathbf{E}_{\theta\phi} = \mathbf{r}_{\phi^*}\otimes\bar{\mathcal{D}}_\theta - \mathbf{r}_\phi\otimes\bar{\mathcal{D}}_{\theta^*}$. Suppose $\mathbf{E}_{\theta\phi} = \mathbf{0}$ and apply it to $\mathbf{X} = \mathbf{e}_b\otimes\mathbf{u}$, a single band b with spatial pattern $\mathbf{u}$: $(\mathbf{r}_{\phi^*})_{:b}\otimes\bar{\mathcal{D}}_\theta\mathbf{u} = (\mathbf{r}_\phi)_{:b}\otimes\bar{\mathcal{D}}_{\theta^*}\mathbf{u}$ for all $\mathbf{u}$. If $\theta\neq\theta^*$, then $\bar{\mathcal{D}}_\theta \neq c\,\bar{\mathcal{D}}_{\theta^*}$ for every scalar c: both kernels sum to one and reflect padding preserves constants, so both operators map a constant image to the same constant, which forces $c=1$ and contradicts injectivity. Two linear

maps of rank greater than one that are not scalar multiples of each other map some $\mathbf{u}$ to linearly independent images, and for this $\mathbf{u}$ the identity forces $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_{\phi})_{:b} = \mathbf{0}$, a contradiction. If $\theta = \theta^*$, the identity gives $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_{\phi})_{:b}$ for every b, so $\phi = \phi^*$. Hence $\mathbf{E}_{\theta\phi} \neq \mathbf{0}$ for every $(\theta, \phi) \neq (\theta^*, \phi^*)$ in $\mathcal{G}$; its null space is a proper subspace of measure zero, and so is the finite union of these null spaces. For every $\mathbf{X}$ outside it, $J(\theta, \phi) > 0$ whenever $(\theta, \phi) \neq (\theta^*, \phi^*)$. □

Noise in the criterion J. With independent noise, $\mathbb{E}\, J(\theta, \phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2 + \frac{N}{s^2}\big(M\varsigma_{\mathrm{M}}^2\|\mathbf{k}_\theta\|^2 + \varsigma_{\mathrm{H}}^2\|\mathbf{r}_\phi\|_F^2\big)$ up to border terms, where $\mathbf{k}_\theta$ is the blur kernel. The noise term decreases with θ (a wider kernel has a smaller norm) and barely depends on ϕ, so noise biases $\hat{\theta}$ upward by an amount that vanishes as the SNR grows.

Proposition 1. Let $\varsigma_{bij} = \alpha_k \lambda_i^{\mathrm{r}} \lambda_j^{\mathrm{c}} + \beta_k \nu_b$, the eigenvalues of $\tilde{\mathbf{A}}^\top \mathbf{W}_k \tilde{\mathbf{A}}$, and e^f the numerator of Γ, so $\frac{1}{2} < e^f < 2$. (a) $\mathbf{G}_k = \mathbf{0}$ at $\mathbf{X}^*$. (b) The step maps a difference $\mathbf{V} - \mathbf{V}'$ to $\mathcal{Q}\big((1-\Gamma\varsigma) \odot \mathcal{Q}^{-1}(\mathbf{V} - \mathbf{V}')\big)$, and $0 \leq \Gamma\varsigma = e^f\varsigma/(1+\varsigma) < 2$, so every factor lies in $(-1, 1]$. (c) The displacement is $\mathcal{Q}\big(\Gamma \odot \mathcal{Q}^{-1}(\mathbf{B}^\top \mathbf{e})\big)$, whose operator norm is max $e^f\sqrt{\varsigma}/(1+\varsigma) \leq 2 \cdot \frac{1}{2} = 1$. □

S3. Data and protocol

Chikusei. The scene [22] has 2517×2335 pixels and 128 bands from 363 to 1018 nm. The split sizes follow [5]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64. Their locations are not specified there, so we fix them: we take the central 2048×2048 crop, which has no no-data border, and divide it by its global maximum (15,133 digital numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048. Sixteen-row gaps keep the regions disjoint.

Pavia Centre. The ROSIS scene [9] has 1096×715 pixels and 102 bands between 430 and 860 nm. As in [5] we keep bands 11–102 (92 bands) and use a four-band multispectral image. The benchmark files distributed with PSRT [6] were not accessible, so we rebuild the benchmark from the original scene to the same specification: division by the maximum of the retained bands (8,000 DN), four 256×256 test tiles in the top-left 512×512 region, 63 validation patches on a 32-pixel grid in rows 528–656, and random training crops from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), with 16-pixel gaps.

CAVE and Harvard. CAVE [21] has 32 indoor scenes of 512×512 pixels and 31 bands (400–700 nm); the first 20 scenes train and the last 12 test, as in MHF-Net, MoG-DCN and DCT. Harvard [4] has 1040×1392 images with 31 bands (420–720 nm); 30 images train and the central 512×512 crops of 20 images test, each image divided by its own maximum. Validation patches come from the bottom-right corners of the training images. Both are paired with a Nikon RGB response.

NEON (real data). The NEON airborne observation platform [11] records an imaging spectrometer (1 m) and an RGB camera (0.1 m) on the same flights. We use eight tiles of the SERC site (2019), 120 spectrometer bands between 400 and 1000 nm, and the camera image resampled to 0.25 m. The camera's spectral response is not published, so we fit a unit-sum nonnegative response of the three camera channels to the spectrometer bands on co-located 4 m blocks (R^2 between 0.45 and 0.79). At reduced resolution, the measured 1 m spectrometer image is the reference, the LR-HSI is simulated from it at $\times 4$, and the measured camera image is the HR-MSI; at full resolution the measured 1 m image is fused with the 0.25 m camera image and evaluated without a reference (D_λ, D_s, HQNR [2]). Table S5 lists the results.

Spectral responses. WorldView-2 (Chikusei) and IKONOS (Pavia Centre) publish their band responses only as plots with band edges. We model each band as a flat top with logistic flanks (4 nm roll-off) that pass through the published edges, at 5% response for WorldView-2 and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Degradation. Every image, including every training crop, is degraded on its own with (1): reflect padding, a normalized Gaussian blur (7×7, $\sigma = 2$ at $\times 4$; $(2s-1) \times (2s-1)$, $\sigma = s/2$ at $\times s$), decimation from index zero, and the spectral response. At $\times 16$ all methods train on 128×128 patches with batch 4.

S4. Baselines

All published networks are built from their authors' repositories, cloned at run time; only the numbers of input and output bands and the scale factor are changed. PSRT: github.com/Deng-shangqi/PSRT (9c2be90); DCT: github.com/qingma2016/DCTransformer (bd75b15); MIMFormer: github.com/meiruni/MIMFormer (0904158); DHIF-Net: github.com/TaoHuang95/DHIF-Net (6434366), whose code covers $\times 8$ and $\times 16$, so at $\times 4$ we use one of the two stride-4 layers of its $\times 16$ operator; SMGU-Net: github.com/yansql/SMGU-Net (115ce77), written for $\times 4$ only; CLSNet: github.com/HengYang01/CLSNet

(6b13371), whose upsampling layer is set to the scale factor; OTPNet: github.com/tsharry2201/OTPNet (7c17964), written for pansharpening, with the HR-MSI in place of the panchromatic image; U2Net: github.com/PSRben/U2Net (32e214c), also written for pansharpening, with its input, upsampling and output layers resized; MIMO-SST: github.com/Freelancefangjian/MIMO-SST (06aacc1), whose final full-resolution output is used. DCT and MIMFormer use gradient checkpointing to fit a 16 GB T4, which does not change what they compute. SSR-NET is our re-implementation of [26].

Under the common recipe every baseline uses its published learning rate with Adam(W) and an L1 loss, batch 8, warm-up and cosine decay over the budget, no weight decay and no gradient clipping. The authors' own recipes (Tab. 3) are read from their code: PSRT uses AdamW (learning rate 10^{-4}, decay 10^{-4}), clipping at 10, batch 32 and a constant learning rate; MIMFormer uses Adam (10^{-4}), batch 16 and step decay at 30, 60 and 90 of 101 epochs, mapped onto the time budget.

S5. PUFormer details

Each prior (Fig. S1) is a three-level U-Net with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and the bottleneck, mirrored decoder levels and two refinement blocks. A block (Fig. S2) follows Restormer [25]: transposed attention (Fig. S3) with 1, 2 and 4 heads on the three levels, followed by a gated depth-wise feed-forward network with expansion 2.66. The three priors hold 3.37–3.38 M parameters each; the network has 10.13 M parameters for Chikusei. Training uses AdamW ($\beta = (0.9, 0.99)$, weight decay 10^{-4}, peak learning rate 3×10^{-4}), a 1000-iteration warm-up, cosine decay to 10^{-3} of the peak at the end of the budget, gradient-norm clipping at 0.1, and an exponential moving average of the weights (decay 0.999). The image path, including all operator applications, runs in 32-bit precision and the priors in 16-bit mixed precision. The weights with the best validation PSNR are tested once, without self-ensemble.

S6. Additional results

Comparison with reported numbers. Table S3 places our longest-trained Chikusei model (two runs, 20.4 h in total) next to the 17 methods tabulated in [5]; Tab. S4 does the same for Pavia Centre. These tables give context, not a ranking: the other rows are copied, our test images follow the published protocol but are not necessarily the same pixels, and our retrained baselines exceed their reported values by up to 3.6 dB (*e.g.* MIMFormer 56.96 dB against 53.36 dB reported). The PSNR values of [5] use a fixed peak: for every row of their Chikusei table, RMSE $\cdot\, 10^{\mathrm{PSNR}/20}$ lies between 16.6k and 17.1k DN, above our peak of 15,133 DN, so our PSNR is conservative by about 0.86 dB on this dataset. RMSE in DN does not depend on the peak and is the most direct comparison.

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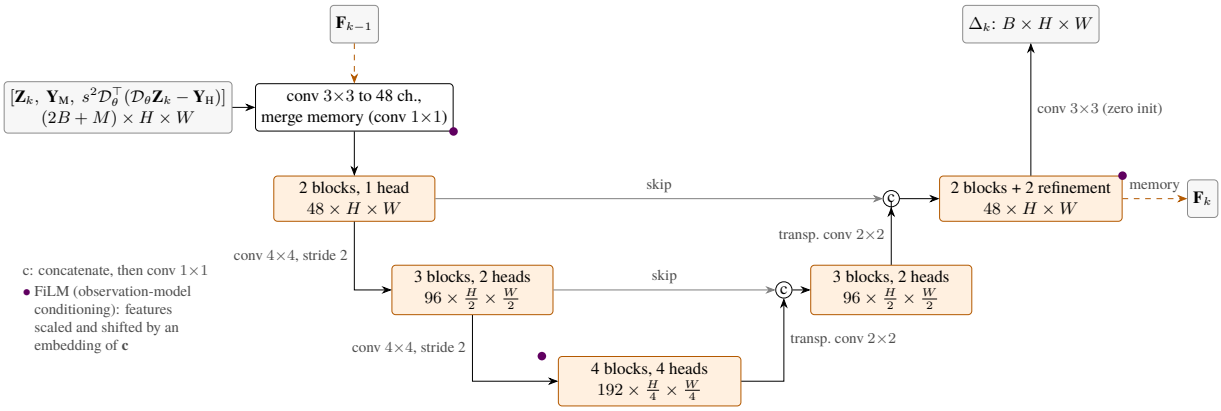


Figure S1. The prior $\mathcal{P}_k$ of stage k (widths for Chikusei, $B = 128$, $M = 8$): a three-level U-Net of transformer blocks (Fig. S2). The encoder halves the resolution twice with strided convolutions, the decoder restores it with transposed convolutions and merges the encoder features through skip connections; the previous stage's decoder features enter after the input convolution, and the last convolution starts at zero so that training starts from the HQS iteration.

Table S1. All metrics of the equal-budget benchmark (single pass; SSIM with the data range set to each image's maximum; RMSE in DN; –: run not available).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
Chikusei ×4								
Bicubic	33.63	0.7404	4.406	9.011	0.8408	0.7648	0.6914	321.81
GSA	43.20	0.9778	2.730	3.279	0.9905	0.9922	0.9596	107.69
SSR-NET	55.55	0.9958	0.954	1.707	0.9948	0.9987	0.9901	25.30
PSRT	53.99	0.9948	1.057	1.794	0.9942	0.9982	0.9887	30.57
DCT	53.18	0.9945	1.133	2.066	0.9930	0.9978	0.9871	33.58
MIMFormer	56.96	0.9964	0.801	1.488	0.9960	0.9989	0.9918	21.65
DHIF-Net	54.39	0.9953	1.020	1.660	0.9951	0.9983	0.9900	29.37
PUFormer	57.95	0.9969	0.731	1.344	0.9966	0.9991	0.9929	19.45
Pavia Centre ×4								
Bicubic	25.52	0.6282	6.554	9.852	0.8174	0.7526	0.6806	436.32
GSA	34.78	0.9438	5.229	3.494	0.9958	0.9922	0.9276	157.93
SSR-NET	48.45	0.9954	2.443	0.758	0.9991	0.9991	0.9719	30.46
PSRT	47.87	0.9958	2.212	0.801	0.9990	0.9989	0.9716	32.60
DCT	46.94	0.9953	2.442	0.888	0.9988	0.9985	0.9712	36.26
MIMFormer	49.63	0.9967	1.982	0.674	0.9993	0.9993	0.9738	26.66
DHIF-Net	47.67	0.9956	2.295	0.840	0.9989	0.9988	0.9704	33.42
PUFormer	50.63	0.9970	1.897	0.621	0.9994	0.9995	0.9754	23.68

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Table S2. All metrics of the equal-budget benchmark on the other datasets and scale factors (single pass; SSIM with the data range set to each image's maximum; RMSE in DN, on the 8-bit scale for CAVE and Harvard; –: run not available).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
CAVE ×4								
Bicubic	30.05	0.8780	4.795	8.229	0.9679	0.7220	0.7518	8.67
GSA	42.30	0.9798	4.651	2.080	0.9981	0.9849	0.8949	2.13
SSR-NET	46.51	0.9942	2.922	1.254	0.9989	0.9938	0.9577	1.29
PSRT	48.51	0.9950	2.394	1.016	0.9993	0.9950	0.9675	1.03
DCT	46.36	0.9934	2.794	1.290	0.9990	0.9928	0.9579	1.31
MIMFormer	50.93	0.9966	1.955	0.786	0.9996	0.9966	0.9718	0.78
DHIF-Net	49.82	0.9962	2.047	0.896	0.9995	0.9959	0.9708	0.90
PUFormer	52.57	0.9973	1.653	0.661	0.9997	0.9976	0.9742	0.65
Harvard ×4								
Bicubic	35.53	0.8588	2.690	4.543	0.9415	0.7894	0.6642	4.69
GSA	46.30	0.9766	2.379	1.669	0.9917	0.9879	0.8695	1.34
SSR-NET	49.18	0.9813	2.148	1.387	0.9920	0.9913	0.8911	0.95
PSRT	49.29	0.9813	2.127	1.344	0.9925	0.9914	0.8921	0.93
DCT	48.92	0.9808	2.188	1.441	0.9917	0.9911	0.8881	0.98
MIMFormer	49.72	0.9822	2.061	1.287	0.9931	0.9923	0.8960	0.88
DHIF-Net	49.57	0.9820	2.084	1.301	0.9930	0.9920	0.8947	0.90
PUFormer	50.01	0.9829	2.012	1.228	0.9937	0.9933	0.9025	0.85
Chikusei ×8								
Bicubic	31.20	0.6503	6.127	5.980	0.7034	0.6287	0.4389	425.27
GSA	38.69	0.9524	3.972	2.572	0.9847	0.9839	0.9069	180.29
SSR-NET	–	–	–	–	–	–	–	–
PSRT	53.05	0.9941	1.172	1.004	0.9927	0.9979	0.9863	33.95
DCT	51.77	0.9934	1.318	1.198	0.9905	0.9973	0.9839	39.69
MIMFormer	55.57	0.9957	0.919	0.862	0.9946	0.9987	0.9897	25.33
DHIF-Net	–	–	–	–	–	–	–	–
PUFormer	56.82	0.9963	0.822	0.759	0.9956	0.9989	0.9912	22.08
Pavia Centre ×8								
Bicubic	23.07	0.4775	8.499	6.541	0.6499	0.6020	0.4060	575.39
GSA	30.24	0.8731	7.163	2.960	0.9927	0.9841	0.8416	269.22
SSR-NET	–	–	–	–	–	–	–	–
PSRT	47.13	0.9953	2.394	0.430	0.9988	0.9987	0.9706	35.49
DCT	45.51	0.9942	2.739	0.521	0.9983	0.9981	0.9697	42.82
MIMFormer	49.09	0.9964	2.064	0.353	0.9992	0.9992	0.9729	28.38
DHIF-Net	–	–	–	–	–	–	–	–
PUFormer	49.94	0.9967	1.986	0.329	0.9993	0.9994	0.9741	25.64
Chikusei ×16								
Bicubic	29.77	0.6186	7.611	3.523	0.5555	0.5163	0.2136	500.32
GSA	35.20	0.9072	5.303	1.860	0.9781	0.9637	0.8093	268.91
SSR-NET	–	–	–	–	–	–	–	–
PSRT	–	–	–	–	–	–	–	–
DCT	–	–	–	–	–	–	–	–
MIMFormer	54.21	0.9951	1.029	0.490	0.9931	0.9984	0.9878	29.54
DHIF-Net	–	–	–	–	–	–	–	–
PUFormer	55.89	0.9960	0.885	0.415	0.9947	0.9988	0.9900	24.43
Pavia Centre ×16								
Bicubic	–	–	–	–	–	–	–	–
GSA	–	–	–	–	–	–	–	–
SSR-NET	–	–	–	–	–	–	–	–
PSRT	–	–	–	–	–	–	–	–
DCT	–	–	–	–	–	–	–	–
MIMFormer	–	–	–	–	–	–	–	–
DHIF-Net	–	–	–	–	–	–	–	–
PUFormer	–	–	–	–	–	–	–	–

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Table S3. Chikusei ×4: PUFormer next to the methods reported in [5] (all other rows copied from there). SSIM of PUFormer as in the PSRT code (data range 1).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [1]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [23]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [18]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [17]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [6]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [16]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [3]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [12]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [7]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNet [24]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [19]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [14]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [8]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [13]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [15]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [20]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [5]	56.1871	0.9982	0.7203	1.5158	**0.9969**	0.9948	0.9891	25.73
PUFormer (single pass)	58.1467	**0.9989**	0.7171	1.3367	0.9930	**0.9967**	**0.9992**	19.06
PUFormer (self-ensemble)	**58.2016**	**0.9989**	**0.7126**	**1.3304**	0.9931	**0.9967**	**0.9992**	**18.94**

Table S4. Pavia Centre ×4 comparison. All rows except PUFormer are reported in [5] on the benchmark data released with PSRT [6]; PUFormer is evaluated on our simulation of the same protocol from the original scene (Section S3). Best value per column in bold. SSIM of PUFormer uses the PSRT definition (data range 1); with data range equal to each image's maximum it is 0.9970 (single pass) and 0.9970 (self-ensemble).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [1]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [23]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [18]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [17]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [6]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [16]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [3]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [12]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [7]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNet [24]	45.8081	0.9967	1.6573	0.9465	**0.9987**	0.9980	0.9968	41.51
SMGU [19]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [14]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [8]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNet [13]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [15]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [20]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [5]	47.1597	0.9972	**1.4542**	0.8262	0.9980	0.9985	0.9976	35.71
PUFormer (single pass)	50.6299	**0.9974**	1.8955	0.6206	0.9751	**0.9994**	**0.9995**	23.68
PUFormer (self-ensemble)	**50.6915**	**0.9974**	1.8833	**0.6170**	0.9753	**0.9994**	**0.9995**	**23.52**

Table S5. Real data (NEON, SERC 2019: imaging spectrometer, 120 bands 400–1000 nm, with the measured camera RGB). Reduced resolution: the LR-HSI is simulated from the 1 m HSI (×4), the RGB is measured and the 1 m HSI is the reference. Full resolution: the measured 1 m HSI is fused with the measured 0.25 m RGB, without a reference. All networks retrained under the same 4 h budget; best retrained network in bold.

	Reduced resolution (reference)				Full resolution (no reference)		
Method	PSNR↑	SAM↓	ERGAS↓	SSIM↑	D_λ↓	D_s↓	HQNR↑
Bicubic	31.78	3.422	13.679	0.7779	0.1966	0.0659	0.7494
GSA	31.25	6.087	14.115	0.7414	0.4259	0.1881	0.4774
SSR-NET [26]	–	–	–	–	–	–	–
PSRT [6]	36.74	2.434	7.483	0.9111	0.1191	0.0514	0.8366
DCT [16]	36.43	2.816	7.841	0.9037	0.1533	0.0666	0.7938
MIMFormer [12]	38.45	2.075	5.893	0.9342	0.0635	0.0465	0.8923
DHIF-Net [10]	36.97	2.357	7.119	0.9129	0.0854	**0.0388**	0.8783
PUFormer	**38.70**	**1.876**	**5.541**	**0.9390**	**0.0273**	0.0585	**0.9156**
zero border	38.68	1.868	5.601	0.9394	0.0278	0.0629	0.9109
learned operator	38.68	1.867	5.562	0.9393	0.0266	0.0651	0.9099
no data step	38.63	1.897	5.640	0.9389	0.0555	0.0569	0.8905
blur + noise randomized	–	–	–	–	–	–	–

Table S6. Ablation: each variant changes one part of the full model and is trained from scratch with the same 4 h budget on one T4 (test set, single pass). Δ: PSNR change against the full model. Cons.: PSNR$(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_{\mathrm{H}})$ in dB. Swap: worst test PSNR over the blurs $\sigma \in \{0.75, 1.25, 1.5\}\sigma_0$ with the true blur given at test time; n/a: no operator to replace. ±: mean and standard deviation over three seeds.

			Pavia Centre ×4				Chikusei ×4			
#	Variant	Params (M)	PSNR	Δ	Cons.	Swap	PSNR	Δ	Cons.	Swap
(1)	PUFormer (full model)	9.99	50.62±0.02	–	77.6	50.50	57.95±0.004	–	83.9	57.70
(2)	no data step, no residual input	9.99	50.46	-0.17	68.9	n/a	57.65	-0.30	73.3	n/a
(3)	no residual input	9.99	50.62	-0.01	75.7	–	58.01	+0.06	82.8	–
(4)	exact kernel, zero border	9.99	50.54±0.03	-0.08	74.5	49.64	57.73±0.05	-0.22	75.9	54.38
(5)	learned blur kernel and SRF	9.99	50.57	-0.06	73.6	n/a	57.73	-0.22	77.1	n/a
(6)	exact proximal step, conjugate gradients	9.99	50.55	-0.07	79.4	50.44	57.74	-0.21	85.2	57.54
(7)	exact proximal step, closed form	9.99	50.65	+0.02	80.9	50.58	57.93	-0.02	87.3	57.76
(8)	learned spectral data step	9.99	50.63	+0.01	82.9	50.50	58.00	+0.05	89.4	57.83
(9)	with the HR-MSI residual input	10.11	50.64	+0.01	79.0	50.58	58.04	+0.09	86.5	57.85
(10)	no cross-stage memory	9.98	50.67	+0.05	77.2	–	57.96	+0.01	84.1	–
(11)	$K = 1$ stage	3.33	50.60	-0.02	71.3	–	57.75	-0.20	77.4	–
(12)	$K = 2$ stages	6.66	50.64	+0.02	75.0	–	57.92	-0.02	81.0	–
(13)	$K = 4$ stages	13.32	50.68	+0.06	80.2	–	57.96	+0.01	85.0	–
(14)	convolutional mixer	9.94	50.60	-0.02	78.5	–	58.02	+0.07	85.2	–
(15)	8×8 window attention	9.99	50.63	+0.01	78.6	–	57.98	+0.03	84.4	–
(16)	L1 + deep supervision only	9.99	50.66	+0.03	77.5	–	57.97	+0.02	82.2	–
(17)	no deep supervision	9.99	50.50	-0.12	77.7	–	57.99	+0.04	84.1	–

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Table S7. Operator ablation across datasets and scale factors (one run each, seed 0, 4 h on one T4; test PSNR in dB). Training blur: PSNR at the blur used for training. Changed blur: worst PSNR over the three test blurs $\sigma \in \{0.75, 1.25, 1.5\}\,\sigma_0$, with the true blur supplied (swap) or the training operator kept (fixed). –: run not available.

	Training blur				Changed blur (worst case)				
Setting	Exact	Zero border	Learned	No data step	Exact, swap	Exact, fixed	Zero border, swap	Learned	No data step
Chikusei ×4	57.95	57.79	57.73	57.65	57.70	51.41	54.38	54.28	–
Pavia Centre ×4	50.63	50.52	50.57	50.46	50.50	43.13	49.64	49.33	–
CAVE ×4	52.57	52.19	52.22	51.69	52.29	48.66	51.82	51.08	51.38
Harvard ×4	50.01	49.97	49.98	49.84	49.95	48.41	49.29	49.69	49.74
Chikusei ×8	56.82	56.53	56.39	–	56.69	50.00	55.27	55.31	–
Pavia Centre ×8	49.94	49.64	49.56	–	49.88	42.69	49.38	48.43	–

Table S8. Operator-randomized training, every test condition (PSNR in dB; columns as in the operator-family table of the main paper). Est. error: mean absolute error of the estimated blur width and SRF shift (nm) over the test images.

		MIMFormer, nominal training	MIMFormer, randomized	PUFormer, nominal training		PUFormer, randomized data only		PUFormer, randomized, zero border	PUFormer, randomized + operator			+ observation-model conditioning		
Condition	Est. error	fixed	fixed	fixed	est.	fixed	est.	est.	fixed	est.	true	fixed	est.	true
Chikusei ×4														
nominal	0.00/0.0	56.96	56.50	57.95	57.95	57.43	57.43	57.46	57.59	57.59	57.59	57.60	57.60	57.60
sigma x0.5	0.00/0.0	49.85	56.16	44.22	54.33	57.11	56.66	57.19	48.71	57.28	57.28	48.94	57.31	57.31
sigma x0.75	0.00/0.0	55.00	56.50	51.41	57.75	57.39	57.30	57.41	54.57	57.56	57.56	54.54	57.57	57.57
sigma x1.25	0.00/0.0	56.24	56.46	54.59	57.86	57.41	57.34	57.44	56.34	57.57	57.57	56.26	57.59	57.59
sigma x1.5	0.00/0.0	55.39	56.41	52.32	57.70	57.39	57.24	57.40	55.08	57.54	57.54	55.00	57.56	57.56
sigma x2	0.00/0.0	54.41	56.35	50.33	57.44	57.34	57.10	57.35	53.76	57.49	57.49	53.66	57.51	57.51
shift -12 nm	0.00/0.0	51.16	54.56	45.52	51.97	57.27	53.44	57.35	48.09	57.44	57.44	45.91	57.50	57.50
shift -8 nm	0.00/0.0	53.47	55.98	49.25	54.22	57.39	55.24	57.44	51.07	57.55	57.55	49.13	57.59	57.59
shift -4 nm	0.00/0.0	55.82	56.41	54.09	56.59	57.43	56.77	57.47	54.88	57.59	57.59	53.65	57.61	57.61
shift +4 nm	0.00/0.0	55.80	56.36	54.28	56.52	57.40	56.72	57.43	54.89	57.56	57.56	53.70	57.57	57.57
shift +8 nm	0.00/0.0	53.69	55.87	50.05	54.24	57.34	55.17	57.37	51.26	57.51	57.51	49.42	57.52	57.52
shift +12 nm	0.00/0.0	51.74	54.82	46.91	52.22	57.25	53.48	57.26	48.52	57.41	57.41	46.47	57.42	57.42
snr 40 dB	0.00/0.0	54.08	54.54	53.13	53.13	55.03	55.03	55.12	55.31	55.31	55.31	55.36	55.36	55.36
snr 35 dB	0.00/0.0	50.93	52.07	49.28	49.28	52.70	52.70	52.87	53.13	53.13	53.13	53.25	53.25	53.25
snr 30 dB	0.00/0.0	46.00	48.40	44.83	44.83	49.33	49.33	49.58	49.93	49.93	49.93	50.14	50.14	50.14
sigma x1.5, shift +8 nm, snr 35 dB	0.00/0.0	49.66	51.79	46.20	48.53	52.63	51.65	52.81	49.29	53.06	53.06	47.63	53.16	53.16
sigma x0.75, shift -8 nm, snr 40 dB	0.00/0.0	51.49	54.10	46.81	51.49	55.00	53.57	55.05	49.74	55.26	55.26	47.86	55.34	55.34
sigma x2, shift +12 nm, snr 30 dB	0.01/0.0	45.78	48.03	42.30	44.35	49.25	48.12	49.49	45.99	49.80	49.80	44.09	49.98	49.98
aniso blur 1.5/0.75 sigma0, 30 deg	–/0.0	55.65	56.49	52.46	52.73	57.26	57.27	56.84	55.42	55.60	57.46	55.28	55.41	57.47
aniso blur 1.25/0.5 sigma0, 0 deg	–/0.0	52.88	56.33	48.32	49.71	56.98	56.74	55.33	51.96	52.55	57.32	52.01	52.69	57.11
disk blur r = 1.25 sigma0	–/0.0	54.16	56.48	49.70	55.17	57.31	57.18	57.00	53.40	56.22	57.44	53.43	56.24	57.34
MSI misregistered 0.5 px	0.02/0.0	40.03	39.41	40.51	40.51	39.43	39.43	39.81	40.13	40.13	–	40.17	40.17	–
MSI misregistered 1 px	0.72/0.1	34.35	33.71	34.85	34.73	33.70	33.70	34.03	34.45	34.36	–	34.45	34.36	–
aniso blur, MSI 0.5 px, snr 35 dB	–/0.0	39.63	39.25	39.79	39.79	39.32	39.32	39.63	39.88	39.88	–	39.83	39.83	–
Pavia Centre ×4														
nominal	0.00/0.0	49.63	49.37	50.63	50.63	50.40	50.40	50.41	50.45	50.45	50.45	50.42	50.42	50.42
sigma x0.5	0.00/0.0	43.13	49.09	35.56	46.87	50.19	49.84	50.26	40.48	50.31	50.31	41.38	50.25	50.25
sigma x0.75	0.00/0.0	47.78	49.35	43.13	50.54	50.38	50.33	50.41	46.83	50.45	50.45	47.13	50.41	50.41
sigma x1.25	0.00/0.0	48.94	49.35	46.73	50.58	50.38	50.36	50.40	48.83	50.43	50.43	48.89	50.40	50.40
sigma x1.5	0.00/0.0	48.16	49.33	44.18	50.50	50.36	50.32	50.38	47.33	50.42	50.42	47.49	50.37	50.37
sigma x2	0.00/0.0	47.26	49.29	42.04	50.34	50.33	50.26	50.35	45.80	50.39	50.39	46.03	50.34	50.34
shift -12 nm	0.00/0.0	47.60	47.14	48.73	48.84	50.19	48.52	50.26	48.75	50.28	50.28	47.77	50.32	50.32
shift -8 nm	0.00/0.0	48.63	48.34	49.70	49.74	50.30	49.42	50.35	49.62	50.38	50.38	49.07	50.37	50.37
shift -4 nm	0.00/0.0	49.35	49.07	50.38	50.38	50.37	50.11	50.40	50.22	50.43	50.43	50.05	50.40	50.40
shift +4 nm	0.00/0.0	49.43	49.19	50.42	50.42	50.40	50.18	50.41	50.26	50.45	50.45	50.04	50.41	50.41
shift +8 nm	0.00/0.0	48.86	48.56	49.83	49.85	50.37	49.51	50.38	49.73	50.42	50.42	49.12	50.39	50.39
shift +12 nm	0.00/0.0	48.06	47.54	49.04	49.09	50.32	48.59	50.31	48.98	50.37	50.37	47.98	50.35	50.35
snr 40 dB	0.00/0.0	47.95	48.12	47.68	47.68	48.71	48.71	48.76	48.87	48.87	48.87	48.91	48.91	48.91
snr 35 dB	0.00/0.0	45.68	46.29	44.56	44.56	46.75	46.75	46.86	47.10	47.10	47.10	47.18	47.18	47.18
snr 30 dB	0.00/0.0	42.10	43.19	40.40	40.40	43.72	43.72	43.91	44.30	44.30	44.30	44.44	44.44	44.44
sigma x1.5, shift +8 nm, snr 35 dB	0.00/0.0	44.82	45.84	41.83	44.41	46.72	46.36	46.85	45.29	47.07	47.07	45.34	47.14	47.14
sigma x0.75, shift -8 nm, snr 40 dB	0.00/0.0	46.28	47.20	42.40	47.24	48.64	48.04	48.71	45.93	48.83	48.83	46.06	48.88	48.88
sigma x2, shift +12 nm, snr 30 dB	0.00/0.0	41.35	42.63	38.32	40.40	43.69	43.29	43.91	42.48	44.27	44.27	42.52	44.39	44.39
aniso blur 1.5/0.75 sigma0, 30 deg	–/0.5	47.68	49.34	42.83	42.99	50.00	50.00	48.70	46.71	46.86	50.43	46.99	47.12	50.37
aniso blur 1.25/0.5 sigma0, 0 deg	–/1.0	46.18	49.30	40.33	42.13	49.88	49.68	47.84	44.29	45.08	50.39	44.82	45.39	50.22
disk blur r = 1.25 sigma0	–/1.0	46.80	49.31	41.28	46.89	50.14	49.99	49.39	45.35	48.38	50.42	45.79	48.54	50.29
MSI misregistered 0.5 px	0.03/2.2	31.75	31.18	32.36	32.37	31.35	31.37	31.65	31.94	31.97	–	31.93	31.97	–
MSI misregistered 1 px	0.85/5.0	26.31	25.69	27.00	26.88	25.81	25.83	26.08	26.58	26.48	–	26.47	26.37	–
aniso blur, MSI 0.5 px, snr 35 dB	–/1.8	31.53	31.11	31.80	31.81	31.34	31.35	31.53	31.75	31.76	–	31.68	31.70	–

Table S9. Where the margin over the retrained networks comes from (4 h on one T4, single pass). The two strongest baselines are retrained with their authors' own recipe and with PUFormer's; PUFormer is reduced to its prior alone, to one stage, to three stages without the data step, and trained with the baselines' recipe. Δ: PSNR against the full PUFormer of the same dataset; it.: training iterations completed.

	Chikusei $\times 4$				Pavia Centre $\times 4$			
Model and training	PSNR	Δ	SAM	it.	PSNR	Δ	SAM	it.
MIMFormer								
pipeline recipe (batch 8, AdamW, no decay, cosine)	56.96	-0.99	0.801	100k	49.63	-1.00	1.982	97k
authors' recipe (batch 16, Adam, step decay)	56.00	-1.95	0.874	68k	49.32	-1.31	2.022	68k
PUFormer's recipe (lr 3×10^{-4}, decay, clipping, SAM+SSIM)	–	–	–	–	–	–	–	–
PSRT								
pipeline recipe (batch 8)	53.99	-3.95	1.057	64k	47.87	-2.76	2.212	65k
authors' recipe (batch 32, AdamW decay, clipping, constant lr)	49.51	-8.44	1.551	17k	45.44	-5.19	2.597	17k
PUFormer								
prior alone: one U-Net, no data step	57.67	-0.28	0.747	113k	50.39	-0.24	1.934	115k
one stage with data step	57.75	-0.20	0.743	97k	50.60	-0.03	1.909	109k
three stages, no data step	57.65	-0.30	0.752	41k	50.46	-0.17	1.928	42k
baselines' recipe (lr 10^{-4}, no decay, no clipping, L1)	–	–	–	–	–	–	–	–
full model	57.95	–	0.731	36k	50.63	–	1.897	37k

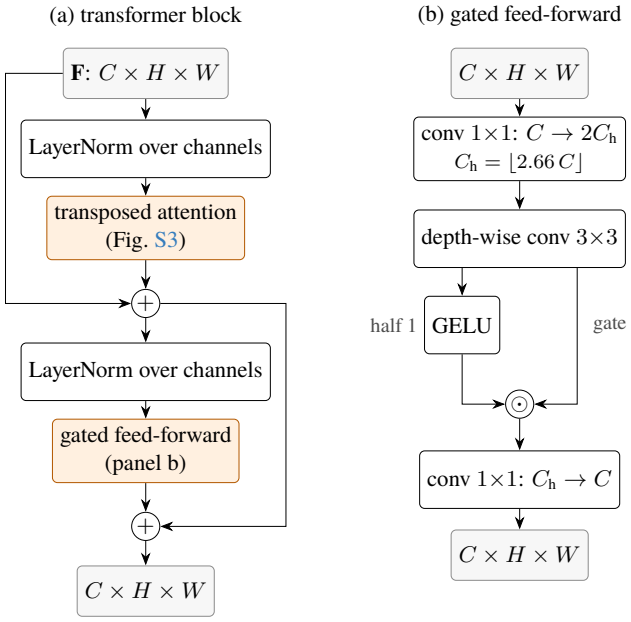


Figure S2. (a) A transformer block of the prior: layer normalization over channels, transposed attention and a gated feed-forward network, each in a residual branch. (b) The gated feed-forward network: the expanded features are split in two halves, and the GELU of one half gates the other.

Table S10. Complexity for one 256×256 Chikusei image (128 bands, $\times 4$) on one Tesla T4.

Method	Params (M)	GFLOPs	Time (ms)	Memory (GB)
PUFormer	10.13	296	178	0.44
SSR-NET	0.44	58	11	0.11
PSRT	0.30	43	118	0.42
DCT	8.46	2640	2103	1.10
MIMFormer	16.46	224	62	0.45
DHIF-Net	48.67	14930	2938	2.08

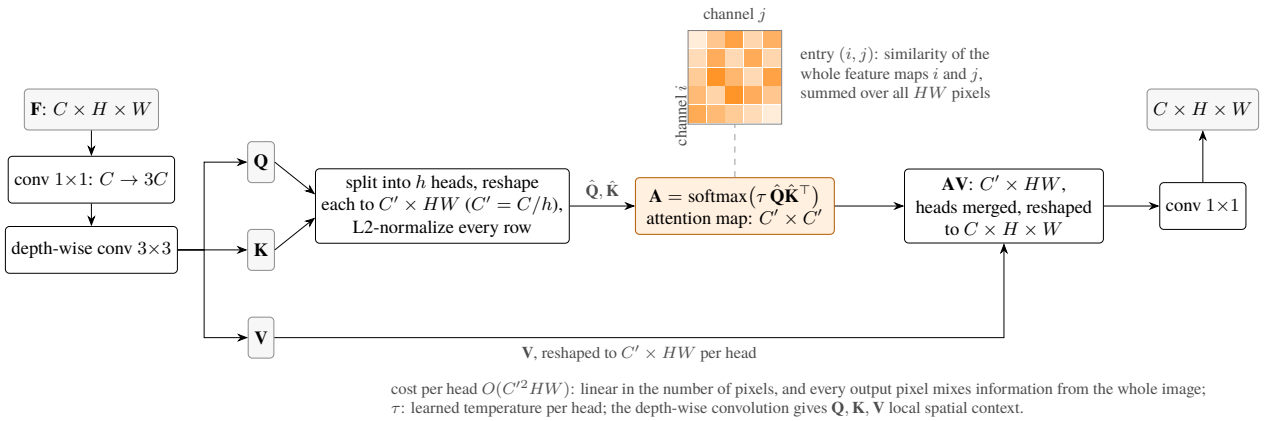


Figure S3. Multi-Dconv-head transposed attention [25], the token mixer of the prior. Queries, keys and values are computed by point-wise and depth-wise convolutions, and the attention map is formed between the C' feature channels of each head rather than between the HW pixels, so its size does not depend on the image size. Replacing it by a depth-wise convolution changes the PSNR by less than 0.1 dB in the ablation.